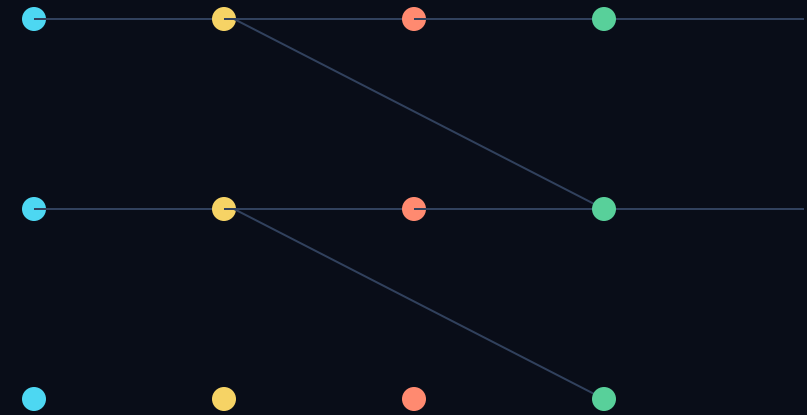


A practical guide for tokens, embeddings, and RAG

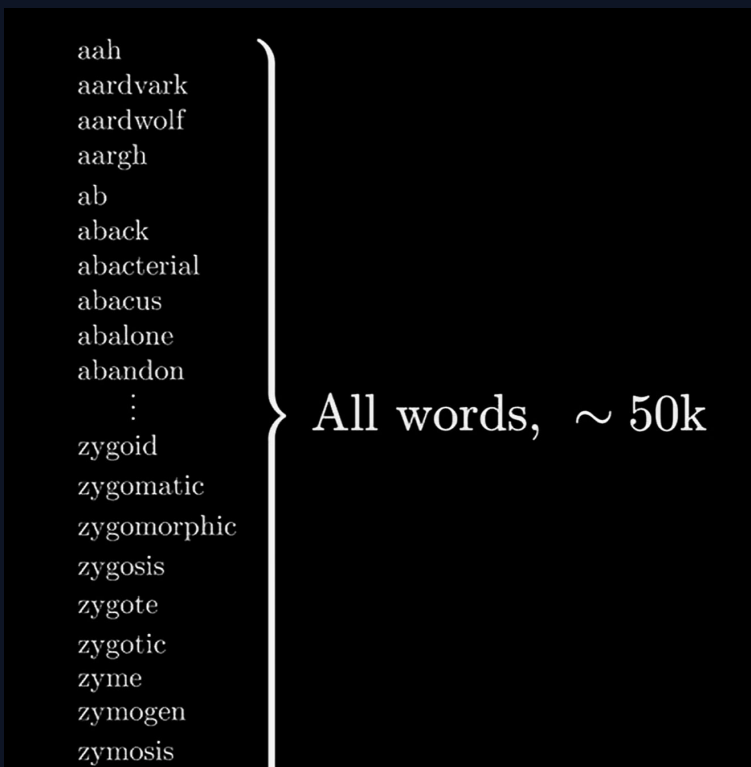
How modern AI turns language into coordinates,
finds related knowledge, and uses it to answer.

Tokens → Vectors → Retrieval →
Answers



Before a model can read, it needs an address book

A vocabulary maps text pieces to stable IDs that a model can look up.



aah
aardvark
aardwolf
aargh
ab
aback
abacterial
abacus
abalone
abandon
⋮
zygoïd
zygomatîc
zygomorphîc
zygoîsîs
zygoîte
zygoîtîc
zyme
zymogen
zymosis

} All words, ~ 50k

Token

A unit the model recognizes and assigns an ID.

Example token lookup

program → token ID 18,427

Embedding Matrix

Columns are vocabulary entries. Each column stores a dense numeric representation.

①

All words, ~ 50k

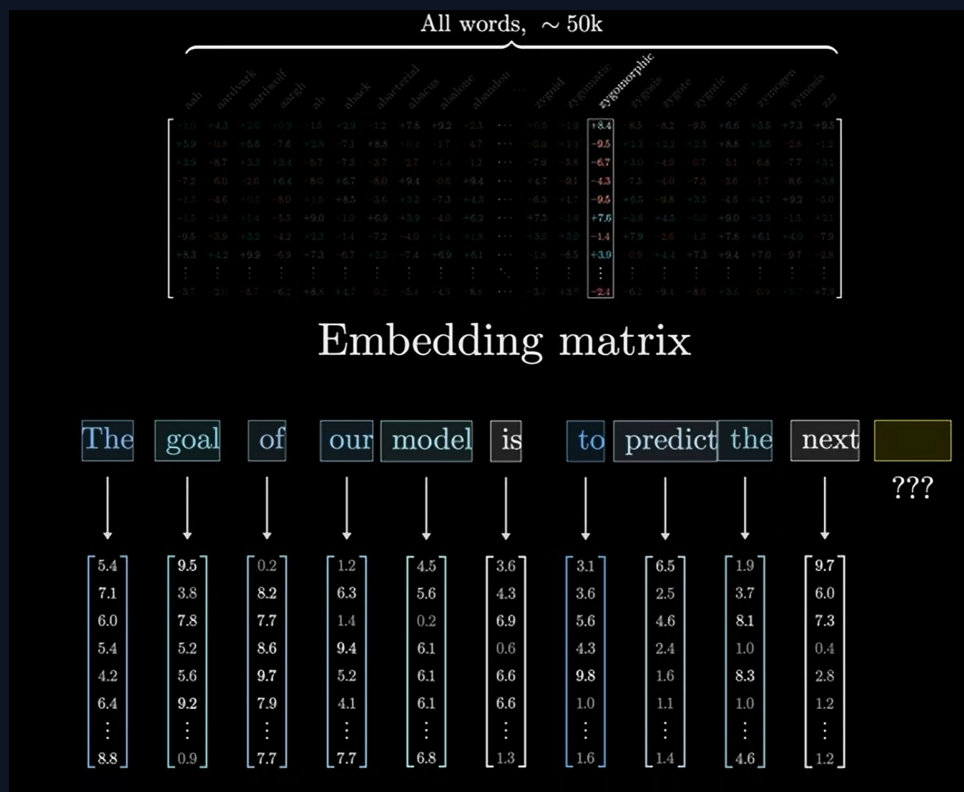
aah	aardvark	aardwolf	aargh	ab	aback	abacterial	abacus	abalone	abandon	...	zygoid	zygomatic	zygomorphic	zygosis	zygote	zygotic	zyne	zymogen	zymosis	zzz
+1.0	+4.3	+2.0	+0.9	-1.5	+2.9	-1.2	+7.8	+9.2	-2.3	...	+0.6	+1.3	+8.4	-8.5	-8.2	-9.5	+6.6	+5.5	+7.3	+9.5
+5.9	-0.8	+5.6	-7.6	+2.8	-7.1	+8.8	+0.4	-1.7	-4.7	...	-0.9	+1.4	-9.5	+2.3	+2.2	+2.3	+8.8	+3.6	-2.8	-1.2
+3.9	-8.7	+3.3	+3.4	-5.7	-7.3	-3.7	-2.7	+1.4	-1.2	...	-7.9	-5.8	-6.7	+3.0	-4.9	-0.7	-5.1	-6.8	-7.7	+3.1
-7.2	-6.0	-2.6	+6.4	-8.0	+6.7	-8.0	+9.4	-0.6	+9.4	...	+4.7	-9.1	-4.3	-7.5	-4.0	-7.5	-3.6	-1.7	-8.6	+3.8
+1.3	-4.6	+0.5	-8.0	+1.5	+8.5	-3.6	+3.3	-7.3	+4.3	...	-6.3	+1.7	-9.5	+6.5	-9.8	+3.5	-4.6	+4.7	+9.2	-5.0
+1.5	+1.8	+1.4	-5.5	+9.0	-1.0	+6.9	+3.9	-4.0	+6.2	...	+7.5	+1.6	+7.6	+3.8	+4.5	+0.0	+9.0	+2.9	-1.5	+2.1
-9.5	-3.9	+3.2	-4.2	+2.3	-1.4	-7.2	-4.0	+1.4	+1.8	...	+3.0	+3.0	-1.4	+7.9	-2.6	-1.3	+7.8	+6.1	+4.0	-7.9
+8.3	+4.2	+9.9	-6.9	+7.3	-6.7	+2.3	-7.4	+6.9	+6.1	...	-1.8	-8.5	+3.9	-0.9	+4.4	+7.3	+9.4	+7.0	-9.7	-2.8
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮
-3.7	-2.0	-5.7	-6.2	+8.8	+4.7	-0.2	-5.4	-4.9	-8.8	...	-3.7	+3.9	-2.4	-6.3	-9.4	-8.6	+3.6	-0.9	+0.7	+7.9

Embedding matrix

≈ 50k vocabulary entries

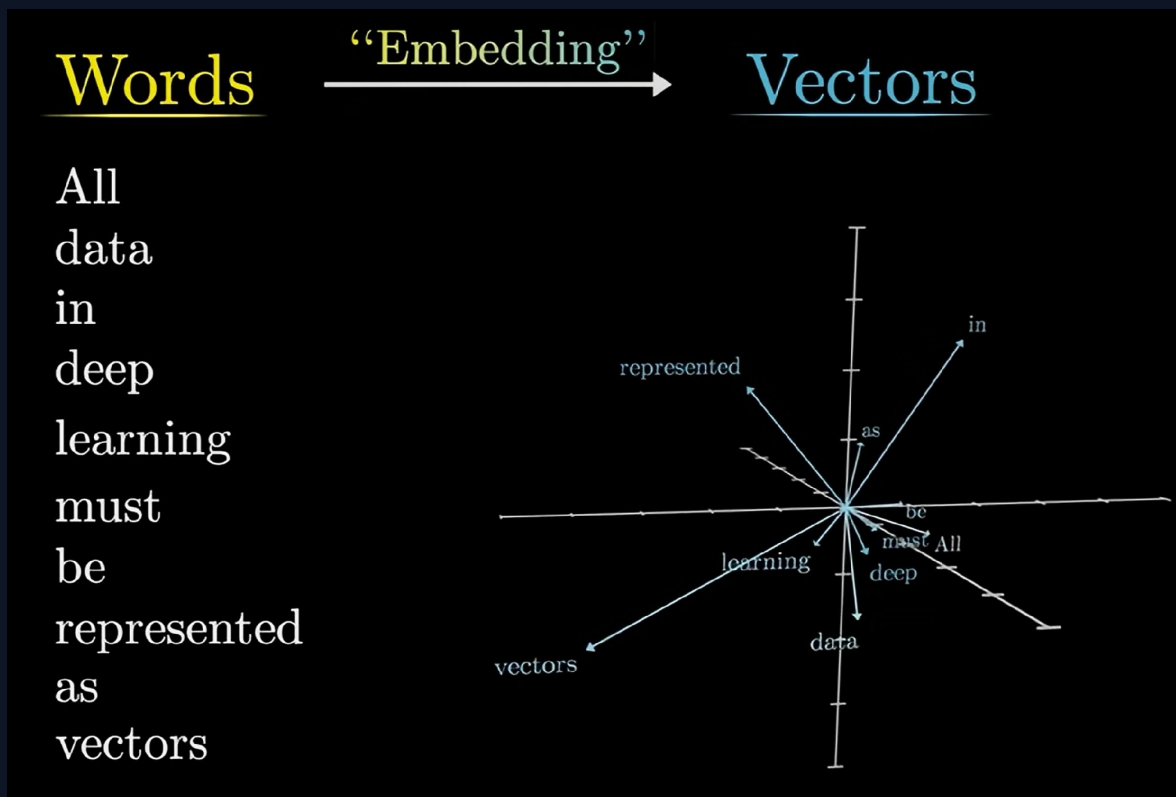
Each token becomes a vector through a lookup

The sequence “The goal of our model...” becomes a sequence of numeric vectors.



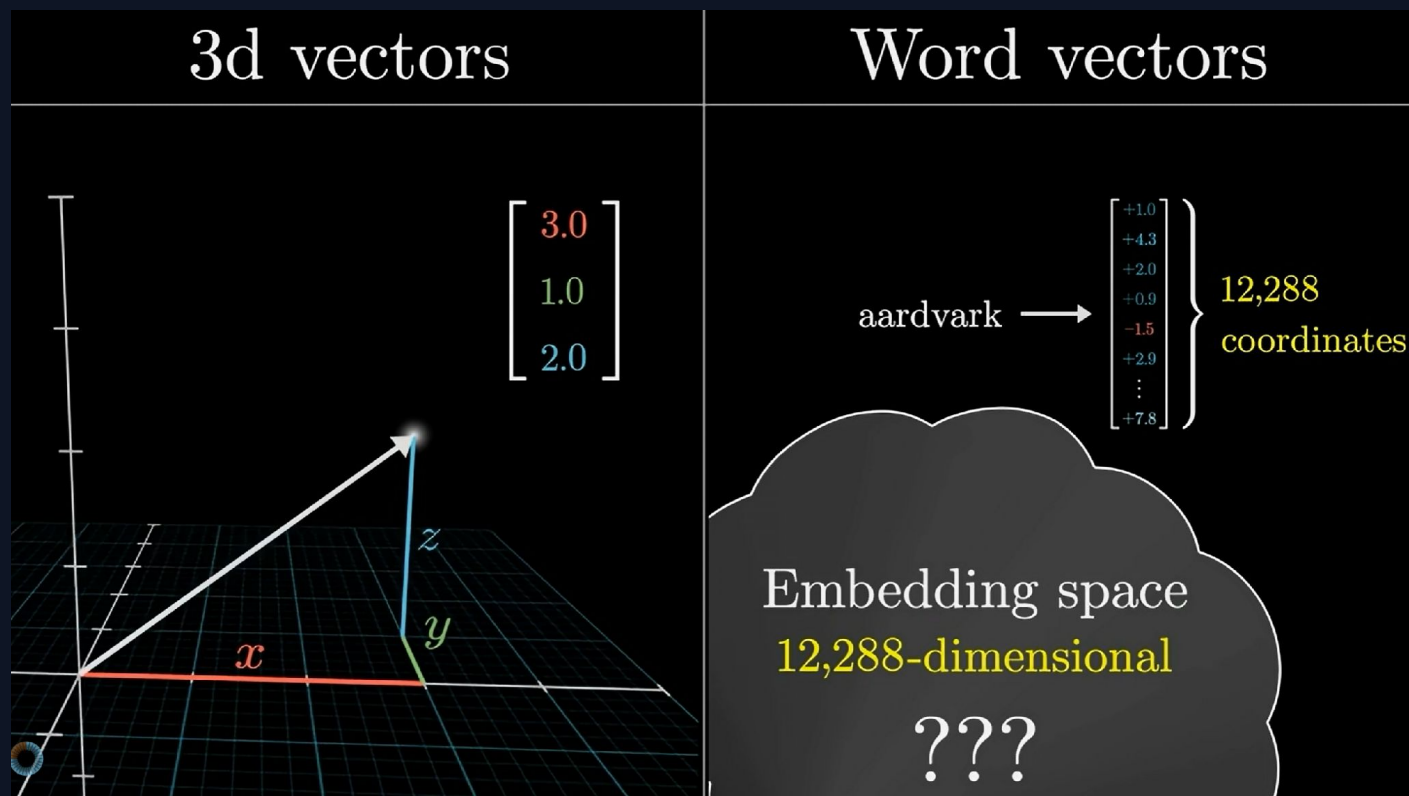
A vector is a location in a space

Words used in similar contexts tend to land near one another.



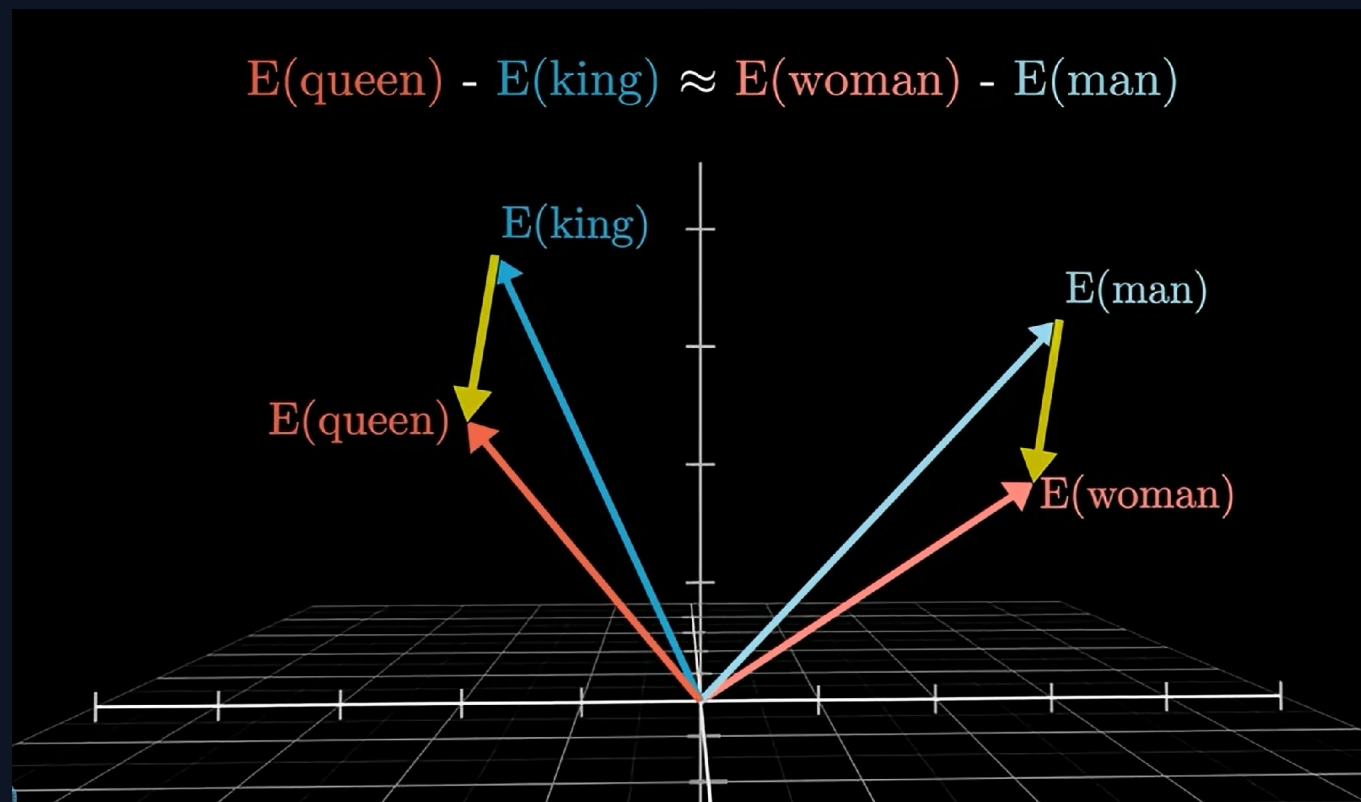
We draw 3D & Real embeddings use thousands of dimensions

More coordinates create room to represent many overlapping patterns at once.

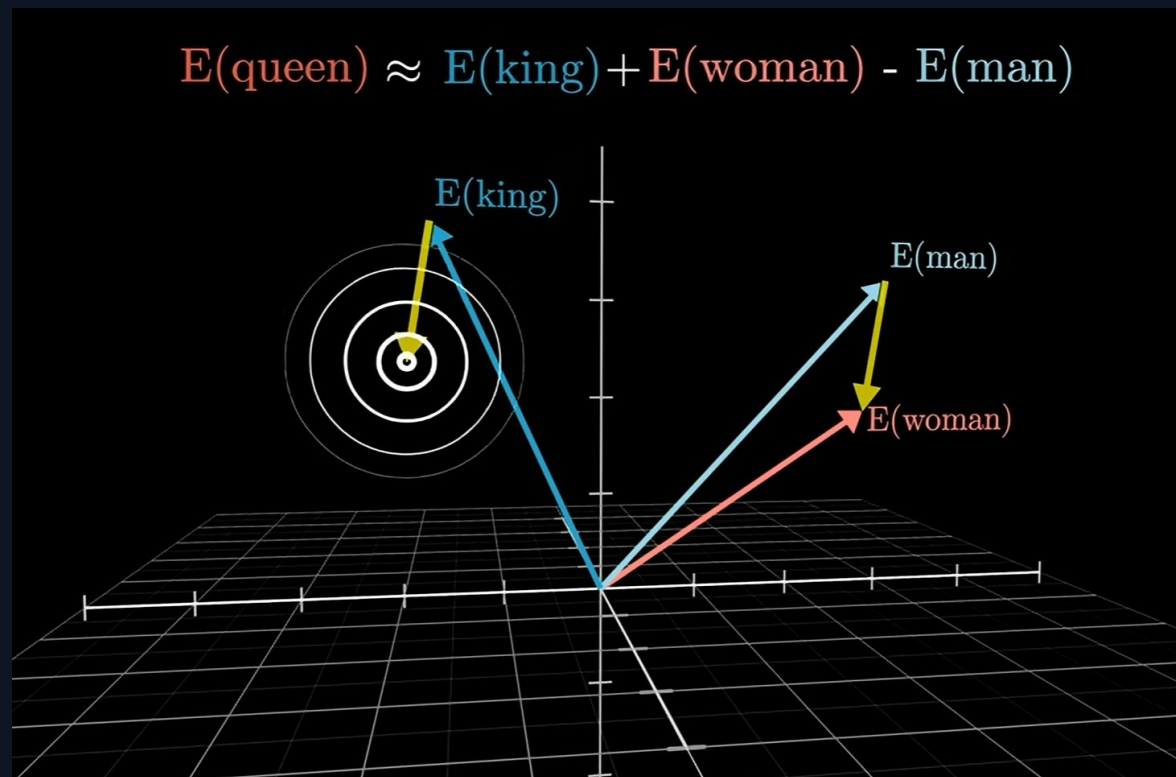


Some relationships show up as consistent directions

The queen-to-king change resembles the woman-to-man change in this classic example.

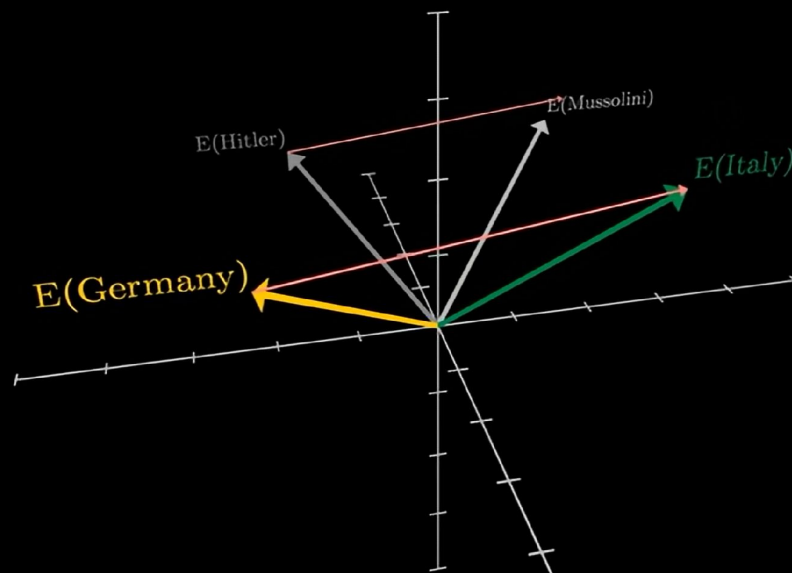


Example 2



Example 2

$$E(\text{Hitler}) + E(\text{Italy}) - E(\text{Germany}) \approx E(\text{Mussolini})$$



5 TB of company knowledge. One question.

An LLM starts with a training snapshot, not a live connection to your internal systems.

YOUR COMPANY

5 TB

Internal
knowledge

- Policies
- Runbooks
- Tickets
- Design docs



Someone asks:

“Can I return
this
after 30 days?”



LLM

He will come up with a
false answer, or in our
language, he will be
hallucinating

The answer exists in the 5 TB. The missing step is finding the right passage at question time.

RAG (Retrieval Augmented Generation)

Retrieval: Finding useful information from an outside source, such as documents, a wiki, or a database.

Augmented: Adding the found information to the model's context.

Generation: Creating an answer with the language model.

RAG has two moments: index knowledge, then retrieve it

Prepare the knowledge once. At question time, retrieve only the best context for the model.

INDEX ONCE Prepare internal knowledge for semantic search



QUERY TIME At question time, retrieve only the most relevant context





Questions?